Construction Chemicals

SUBMITTAL DOCUMENT

# Mixflex PU Binder

One-Component Polyurethane
Binder for Rubber Granules




**Solvent-Free**
100 % solids

**Elastic**
Durable bond

**Working**
30 – 60 min

**1K**
Moisture curing

| | |
|---|---|
| **Project** | |
| **Main Contractor** | |
| **Consultant** | |
| **Submittal No. / Rev.** | 00 |
| **Date** | Nov 2026 |

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METHOD STATEMENT

# Mixflex PU Binder — One-Component Polyurethane Binder for Rubber Granules

| | | | |
|---|---|---|---|
| **Project Name** | | **Document No.** | OMX-MS-MIXFLEXPUBINDER |
| **Main Contractor** | | **Revision** | 01 |
| **Consultant** | | **Date of Issue** | Nov 2026 |
| **Product** | Mixflex PU Binder | **Status** | For Approval |

## 1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Mixflex PU Binder on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

## 2. Scope of Work

This method statement covers preparation, mixing, application, curing and inspection of Mixflex PU Binder (one-component polyurethane binder for rubber granules).

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Priming of the prepared surface.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

## 3. Definitions

| Term | Meaning |
|---|---|
| **Pot life** | Time the mixed material remains workable after Part A and Part B are combined. |
| **Recoat interval** | Minimum and maximum time between coats to achieve inter-coat adhesion. |
| **WFT / DFT** | Wet film thickness measured during application / dry film thickness after cure. |
| **Dew point** | Temperature at which moisture condenses; the substrate must be at least 3 °C above it. |
| **Kit / set** | Pre-weighed Part A and Part B supplied to be mixed completely. |

## 4. Roles & Responsibilities

| Role | Responsibility |
|---|---|
| **Project Manager** | Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor. |

| Role | Responsibility |
|---|---|
| **Site Engineer** | Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records. |
| **QA/QC Inspector** | Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist. |
| **HSE Officer** | Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise. |
| **Foreman / Applicators** | Carry out the work as per this method statement and report any deviation immediately. |
| **Storekeeper** | Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers. |

## 5. Materials

| Material | Description |
|---|---|
| **Mixflex PU Binder** | 5 kg and 20 kg metal pails; 200 – 220 kg drums. |

### 5.1 Key product data

| Property | Typical Value |
|---|---|
| **Type** | One-component, moisture-curing PU |
| **Colour** | Clear / amber |
| **Solids content** | 100 % |
| **Density** | 1.02 – 1.10 kg/L |
| **Viscosity** | 3000 – 6000 mPa·s |
| **Working time** | 30 – 60 min |
| **Setting** | 8 – 12 hours |
| **Full cure** | 24 hours |

## 6. Tools & Equipment

- Grinder / shot blaster or blasting unit
- Industrial vacuum
- Slow-speed mixer
- Rollers
- Brushes
- Airless spray where specified
- Wet-film comb
- DFT gauge
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Wet-film thickness gauge

- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

### 6.1 Personal protective equipment

- Chemical-resistant gloves
- Goggles
- Overalls
- Respirator where required

## 7. Storage & Handling

- 12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep Part A and Part B of each kit together and do not mix parts from different batches.
- Store away from heat, sparks and ignition sources.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

## 8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

## 9. Surface Preparation

1. Check the substrate: concrete ≥ 28 days, sound and dry; steel prepared to the specified grade.
2. Remove laitance, oil, rust, mill scale and loose coatings mechanically.
3. Remove all dust and contaminants.
4. Measure substrate temperature, air temperature and relative humidity; confirm the substrate is at least 3 °C above dew point.
5. Protect adjacent surfaces, drains and fixtures with masking tape and sheeting.
6. Obtain QA/QC approval of the prepared surface before starting the next step.

## 10. Mixing

1. Condition materials to between +15 °C and +25 °C before mixing.
2. Check the batch numbers and expiry dates of both parts and record them.
3. Weigh binder at 15 – 20 % of mix weight.
4. Mix with granules 3 – 5 minutes.
5. Mix only the quantity that can be applied within the pot life.

## 11. Application

1. Mix granules and binder 3 – 5 minutes.
2. Lay, level and compact within the working time.
3. Work in manageable bays, keeping a wet edge to avoid lap marks.
4. Check consumption by area and wet film thickness regularly during application and record the results.

## 12. Curing & Protection

- Setting 8 – 12 hours; full cure 24 hours.
- Protect the applied area from water, dust, other trades and traffic until the stated cure times are reached.
- Barricade the area and display warning signs.

## 13. Quality Control & Inspection

| Check | Stage | Acceptance Criteria | By |
|---|---|---|---|
| **Substrate** | Before application | Prepared, sound and clean as specified | QA/QC |
| **Climate** | During work | Ambient temperature: +10 °C to +35 °C | Site Engineer |
| **Consumption** | During application | 15 – 20 % of mix weight | QA/QC |
| **Finish** | After curing | Uniform, no defects | QA/QC |

### 13.1 Inspection and test plan

| Activity | Inspection / Test | Frequency | Record |
|---|---|---|---|
| **Material receipt** | Product, batch, expiry, packaging condition | Each delivery | Material inspection report |
| **Climate** | Air / substrate temperature, RH, dew point | Start of shift and every 4 hours | Daily climate log |
| **Surface preparation** | Visual, moisture, soundness | Each area | Inspection request (hold point) |
| **Application** | Consumption, thickness, appearance | Each coat / area | Daily progress report |
| **Final inspection** | Visual, continuity, finish | Each completed area | Handover checklist |

## 14. Defects & Remedies

| Defect | Likely Cause | Remedy |
|---|---|---|
| **Pinholes / bubbles** | Outgassing from porous substrate; air entrapped in mixing | Seal with primer; mix at low speed; de-aerate with spiked roller; sand and recoat |
| **Blushing / tacky surface** | High humidity or low temperature during cure | Wash and abrade; recoat in suitable conditions |
| **Poor adhesion** | Contaminated, weak or damp substrate; recoat window exceeded | Remove, re-prepare surface and reapply |
| **Soft spots** | Incorrect mixing or ratio | Remove affected area and reapply with full kits |
| **Uneven colour / lap marks** | Loss of wet edge; different batches | Abrade and apply a further coat from one batch |

## 15. Health, Safety & Environment

| Hazard | Control Measures |
|---|---|
| **Skin and eye contact** | Gloves, goggles, overalls; eye-wash available |
| **Inhalation of vapours / dust** | Ventilation; masks as required |
| **Slips and trips** | Housekeeping; barricades and signs |
| **Manual handling** | Correct lifting technique; mechanical aids |
| **Electrical tools** | Inspected tools; protected cables |

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Contains isocyanates; ensure ventilation.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.

### 15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

## 16. Cleaning

- Clean tools with the recommended thinner before the material cures.

## 17. Limitations

- Ambient temperature: +10 °C to +35 °C.
- No rain during laying.
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

## Appendix A — Inspection Checklist

| Inspection Item | Result | Remarks |
|---|---|---|
| **Substrate prepared and approved** | ☐ Yes ☐ No | |
| **Batch numbers recorded** | ☐ Yes ☐ No | |
| **Material mixed as specified** | ☐ Yes ☐ No | |
| **Applied at specified consumption** | ☐ Yes ☐ No | |
| **Curing and protection carried out** | ☐ Yes ☐ No | |
| **Final inspection passed** | ☐ Yes ☐ No | |

| Inspected by | Signature | Date |
|---|---|---|
| | | |

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# Technical Data Sheet

Mixflex PU Binder — Submittal Document

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**Product Datasheet**
Edition: 01 | Date: Nov 2026
Sport Court Systems




# Mixflex PU Binder

One-Component Polyurethane Binder for Rubber Granules

## Product Description

Mixflex PU Binder is a one-component, solvent-free polyurethane binder for SBR and EPDM rubber granules. It cures with moisture to form elastic, durable sports and play surfaces.

## Uses

- Binding EPDM and SBR granules.
- Playgrounds and running tracks.
- Shock pads.
- Multi-purpose courts.

## Advantages

- Solvent-free, 100 % solids.
- Elastic, durable bond.
- Working time 30 – 60 min.
- Service -30 °C to +80 °C.
- One component.

## Application

1. **Preparation**
   Weigh granules and binder.
2. **Mixing**
   Add 15 – 20 % binder by weight of the mix and mix 3 – 5 minutes.
3. **Laying**
   Lay within 30 – 60 minutes; level and compact.

## Consumption

**15 – 20 %**
of mix weight

## Packaging

5 kg and 20 kg metal pails; 200 – 220 kg drums.

## Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

## Technical Data

*Typical values at 25 °C*

| Property | Typical Value | Test Method |
|---|---|---|
| **Type** | One-component, moisture-curing PU | — |
| **Colour** | Clear / amber | — |
| **Solids content** | 100 % | — |
| **Density** | 1.02 – 1.10 kg/L | — |
| **Viscosity** | 3000 – 6000 mPa·s | — |
| **Working time** | 30 – 60 min | — |
| **Setting** | 8 – 12 hours | — |
| **Full cure** | 24 hours | — |
| **Service temperature** | -30 °C to +80 °C | — |

## Consumption

| Use | Binder dosage |
|---|---|
| **EPDM wearing layer** | 15 – 20 % |
| **SBR base / shock pad** | 15 – 20 % |

*Consumption is theoretical and varies with substrate porosity, profile and application method.*

## Application Conditions

- Ambient temperature: +10 °C to +35 °C.
- No rain during laying.
- Base dry and primed.

## Application Notes

- Keep containers closed (moisture-curing).
- Lay within the working time.
- Do not lay in rain.

## Health & Safety

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured resin may cause skin sensitisation; avoid skin and eye contact.
- Contains isocyanates; ensure ventilation.

## First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools with the recommended thinner before the material cures.

*The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.*

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# Test Reports

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*To be attached: laboratory test reports supporting the values in the Technical Data Sheet.*

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# Approvals & Certificates

Mixflex PU Binder — Submittal Document

*To be attached: approvals and certificates, where available.*